W6 COS30045

Lecture

Agenda

- 6.1 Histogram
- 6.2 Interaction
  - Filtering
  - Tooltips
  - Animations

Class

Agenda

- D3 Interactions

W6.1 Histogram

Lecture

Agenda

- Process of constructing Histogram
- Loading data
- Generate bins
- Create scales
- Append rectangles
- Add axes

Visualising distributions – Histograms

How often does data occur?

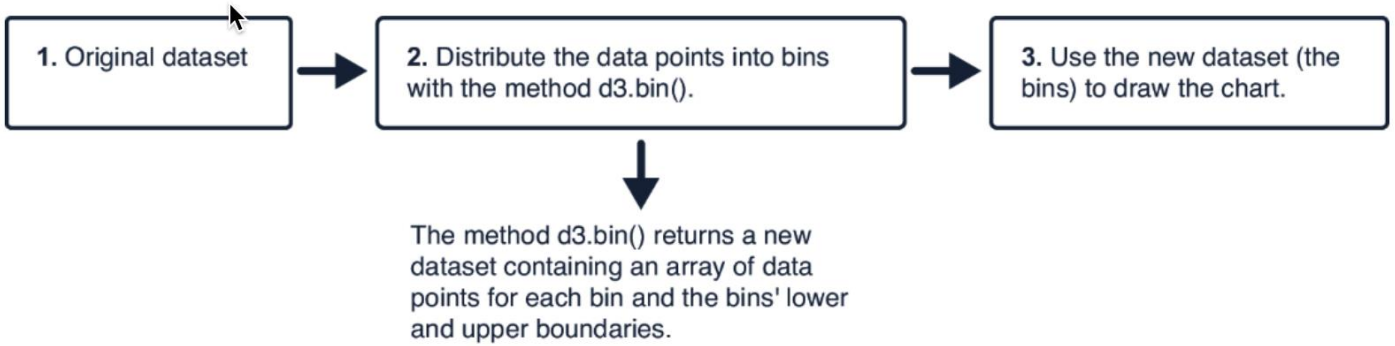
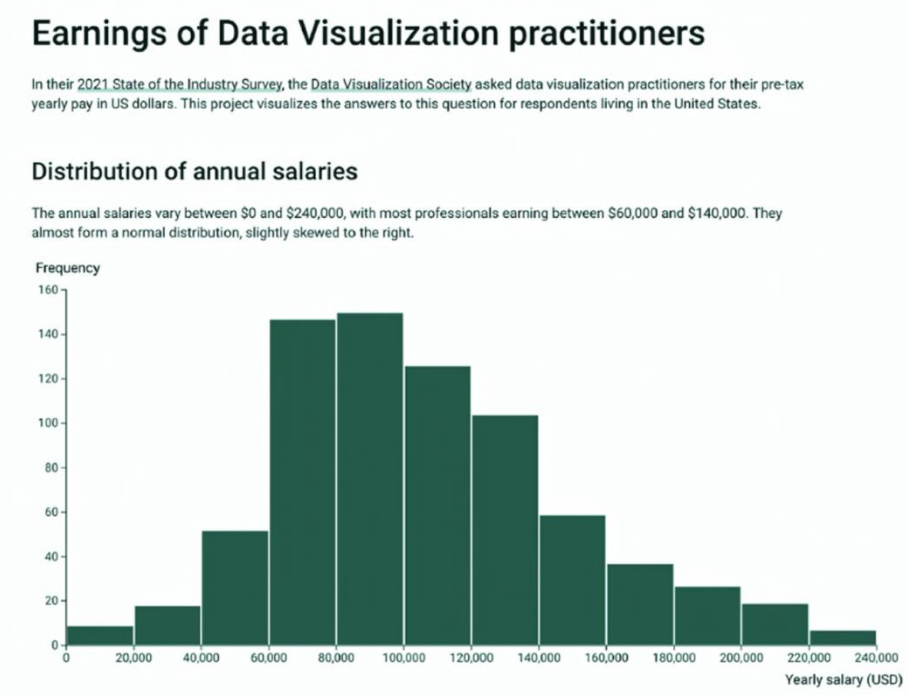


Figure 6.2 To visualize a distribution with D3, we pass the original dataset to function `d3.bin()`. This function returns a new dataset that includes the bins, their boundaries, and their corresponding data points.



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Data for today's chart – average daily screen time for children

age *Number (Integer)*	gender *String*	screen_time_t... *String*	day_type *String*	average_scree... *Number (Float)*	sample_size *Number (Integer)*
5	male	educational	weekday	0.44	500
5	male	recreational	weekday	1.11	500
5	male	educational	weekend	0.5	500
5	male	recreational	weekend	1.44	500
5	female	educational	weekday	0.49	500
5	female	recreational	weekday	0.96	500
5	female	educational	weekend	0.5	500
5	female	recreational	weekend	1.4	500
5	other/prefer not to s	educational	weekday	0.52	500
5	other/prefer not to s	recreational	weekday	0.98	500
5	other/prefer not to s	educational	weekend	0.44	500
5	other/prefer not to s	recreational	weekend	1.51	500
6	male	educational	weekday	0.64	480
6	male	recreational	weekday	1.39	480
6	male	educational	weekend	0.74	480

https://www.kaggle.com/datasets/ak0212/average-daily-screen-time-for-children/data

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Load data

GitHub Copilot:

Please write a d3 data loading function for screen_time.csv data set with headings Age,Gender, Screen_Time_Type, Day_Type, Average_Screen_Time, Sample_Size

```
// Load data
d3.csv("./data/screen_timev2.csv", d => {
  return {
    age: +d.age, // Convert Age to a number
    gender: d.gender.toLowerCase().replace(/\s+/g, "_").replace(/\//g, "_"),
    screen_time_type: d.screen_time_type,
    day_type: d.day_type,
    average_screen_time: +d.average_screen_time, // Convert to a number
    sample_Size: +d.sample_size // Convert to a number
  };
}).then(data => {
  console.log("Processed Data:", data);
  console.log("Loaded Data:", data.map(d => d.gender));

  drawHistogram(data);
  populateFilters(data);
  // populateDaysFilters(data);
```

```
1   // Load data
2   d3.csv("./data/screen_time.csv", d => {
3     return {
4       Age: +d.Age, // Convert Age to a number
5       Gender: d.Gender,
6       ScreenTimeType: d.Screen_Time_Type,
7       DayType: d.Day_Type,
8       AverageScreenTime: +d.Average_Screen_Time, // Convert to a number
9       SampleSize: +d.Sample_Size // Convert to a number
10    };
11  }).then(data => {
12    console.log("Processed Data:", data);
13
14    // Call your visualization function with the processed data
15    drawHistogram(data);
16  });
```

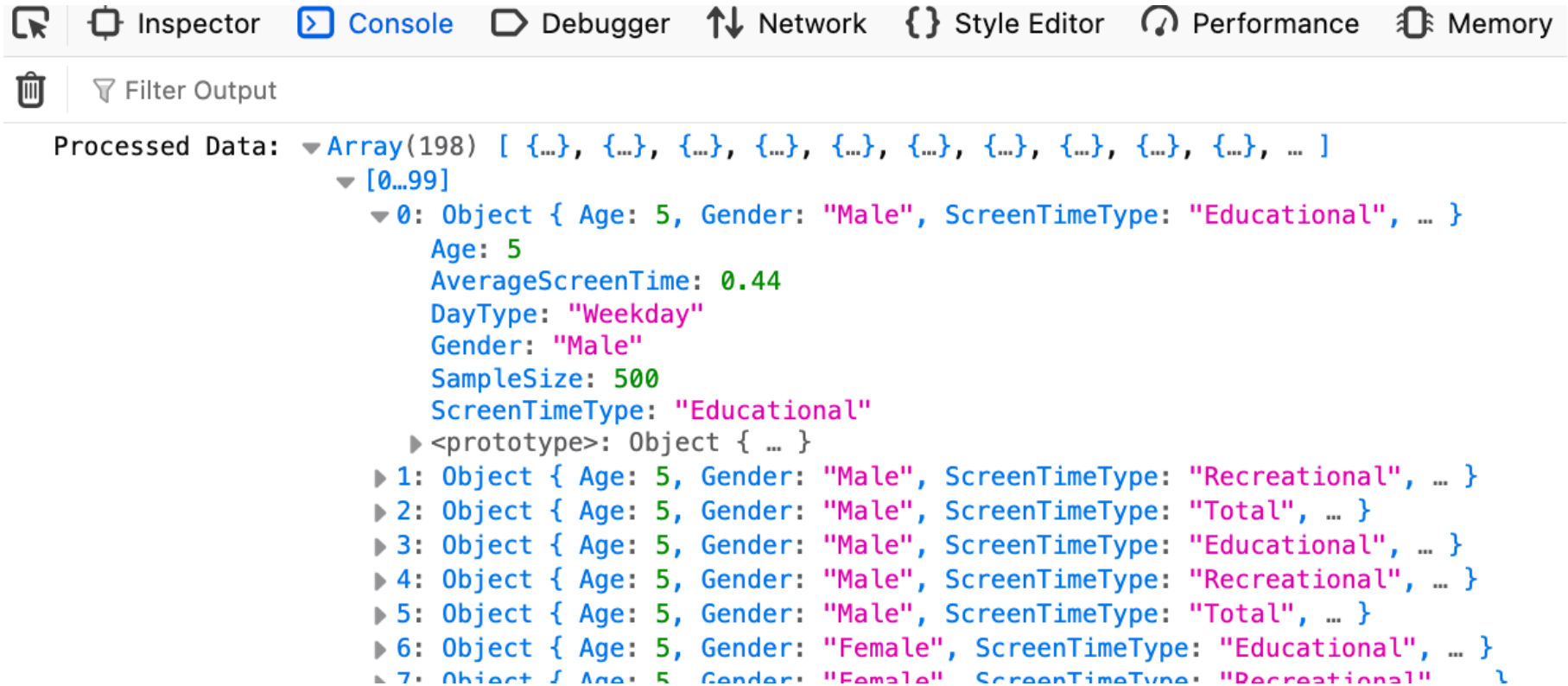
Note if your column headings have spaces in them, it will use a different format to access data

```
d3.csv("./data/screen_time.csv", d => {
  return {
    Age: +d.Age, // Convert Age to a number
    Gender: d.Gender,
    ScreenTimeType: d["Screen Time Type"],
    DayType: d["Day Type"],
    AverageScreenTime: +d["Average Screen Time (hours)"],
    SampleSize: +d["Sample Size"] // Convert to a number
  };
});
```

To keep things neat, and to use same data in different visualisations, consider having a separate file for loading data, just make sure you call it in index.html and load the data to the chart drawing functions you need to create.

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Check Console to see data has been read in



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css files

From Dufour & Meeks (2024) Ch 6

- base.css (styling for heading, cols and rows etc)
- visualization.css

```css
/* Responsive container of the svg element */
.responsive-svg-container {
  width: 100%;
  max-width: 1000px;
}
@media screen and (max-width: 596px) {
  .responsive-svg-container {
    margin-bottom: 20px;
  }
}

/* Axis and labels styles */
.tick text {
  font-family: 'Roboto', sans-serif;
  font-size: 14px;
}
```


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Histogram constants

For controlling where chart placed

Help make it responsive

JS histogram.js > ...

```
const drawHistogram = (data) => {

  /*********************************/
  /*    Declare the constants    */
  /*********************************/
  const margin = {top: 40, right: 30, bottom: 50, left: 40};
  const width = 1000;
  const height = 500;
  const innerWidth = width - margin.left - margin.right;
  const innerHeight = height - margin.top - margin.bottom;


  /*********************************/
  /*    Append the containers    */
  /*********************************/
  // Append the SVG container
  const svg = d3.select("#histogram")
    .append("svg")
      .attr("viewBox", `0, 0, ${width}, ${height}`);

  // Append the group that will contain the inner chart
  const innerChart = svg
    .append("g")
      .attr("transform", `translate(${margin.left}, ${margin.top})`);


};
```

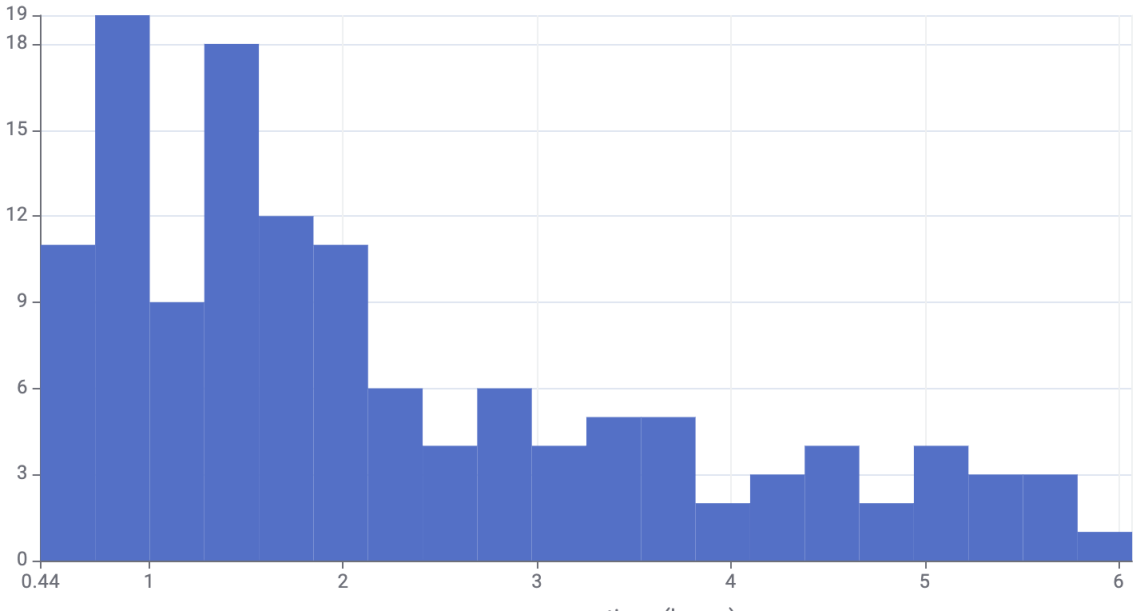

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Generate bins

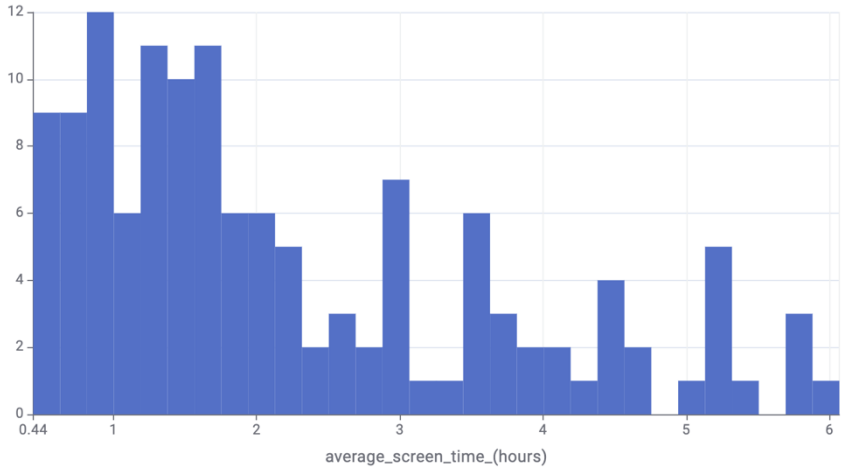
Bin size will give you different granularity

KNIME default bin number for this data: 20

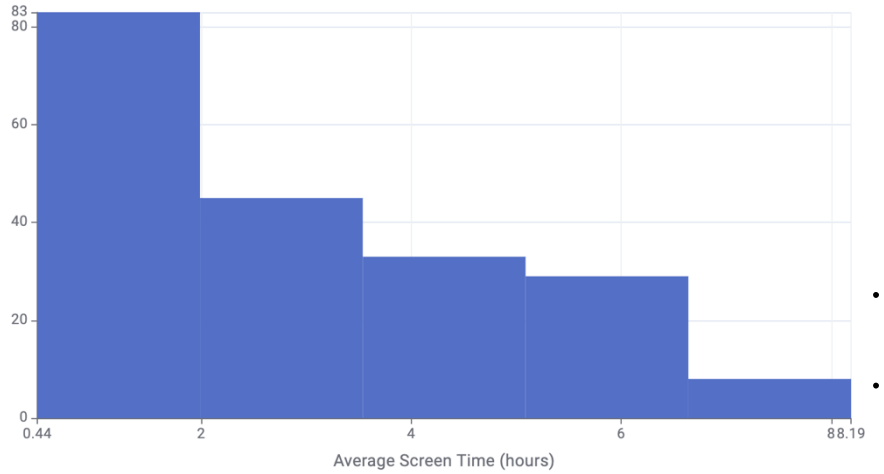
20 bins



30 bins



5 bins



Dufour & Meeks (2024) Ch 6

Organise data into equally sized bins

Check console for bins

- 9 arrays, each is a bin

```
const binGenerator = d3.bin() // Create a bin generator
.value(d => d.AverageScreenTime);

const bins = binGenerator(data); //save the bins into an array
console.log(bins); // Log the bins to the console for debugging
```

```
Processed Data: ▸ Array(198) [ {…}, {…}, {…}, {…}, {…}, {…}, {…}, {…}, {…}, {…}, … ]
▾ Array(9) [ (29) […], (54) […], (31) […], (25) […], (21) […], (21) […], (10) […], (5) […], (2) […] ]
  ▸ 0: Array(29) [ {…}, {…}, {…}, … ]
  ▸ 1: Array(54) [ {…}, {…}, {…}, … ]
  ▸ 2: Array(31) [ {…}, {…}, {…}, … ]
  ▸ 3: Array(25) [ {…}, {…}, {…}, … ]
  ▸ 4: Array(21) [ {…}, {…}, {…}, … ]
  ▸ 5: Array(21) [ {…}, {…}, {…}, … ]
  ▸ 6: Array(10) [ {…}, {…}, {…}, … ]
  ▸ 7: Array(5) [ {…}, {…}, {…}, … ]
  ▸ 8: Array [ {…}, {…} ]
    length: 9
  ▸ <prototype>: Array []
```

```
▾ 7: Array(5) [ {…}, {…}, {…}, … ]
  ▾ 0: Object { Age: 13, Gender: "Other/Prefer not to say", ScreenTimeType: "Total", … }
      Age: 13
      AverageScreenTime: 7.08
      DayType: "Weekend"
      Gender: "Other/Prefer not to say"
      SampleSize: 340
      ScreenTimeType: "Total"
    ▸ <prototype>: Object { … }
  ▸ 1: Object { Age: 14, Gender: "Male", ScreenTimeType: "Total", … }
  ▸ 2: Object { Age: 14, Gender: "Female", ScreenTimeType: "Total", … }
  ▸ 3: Object { Age: 14, Gender: "Other/Prefer not to say", ScreenTimeType: "Total", … }
  ▸ 4: Object { Age: 15, Gender: "Other/Prefer not to say", ScreenTimeType: "Total", … }
    length: 5
    x0: 7
    x1: 8
```

Lower and upper boundary of bin

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Visual of bins (normal distribution)

```
d3.bin()
  .value( d => d.key );
```

Key under which the values to bin are stored

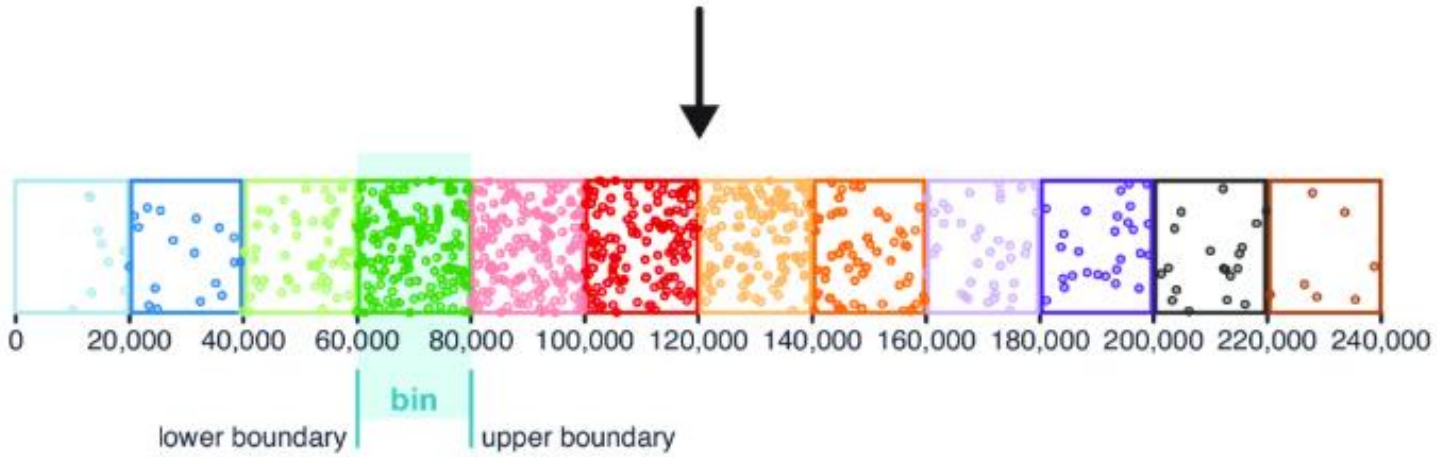
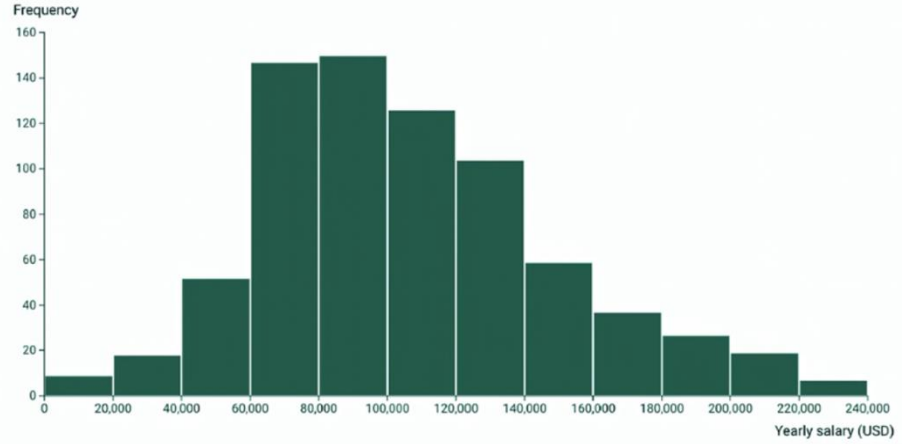


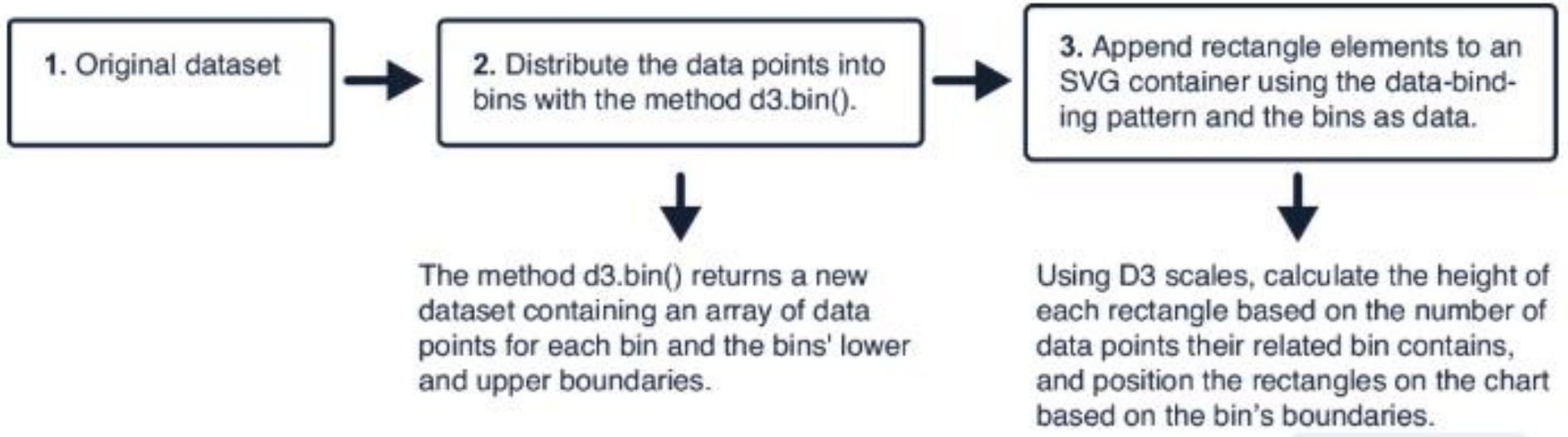
Figure 6.4 Salary data points in their respective bins. The horizontal position of each circle corresponds to the salary, while their vertical positions are arbitrary to reduce overlap.



Dufour & Meeks (2024) Ch 6

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Process of constructing a histogram



Dufour & Meeks (2024) Ch 6

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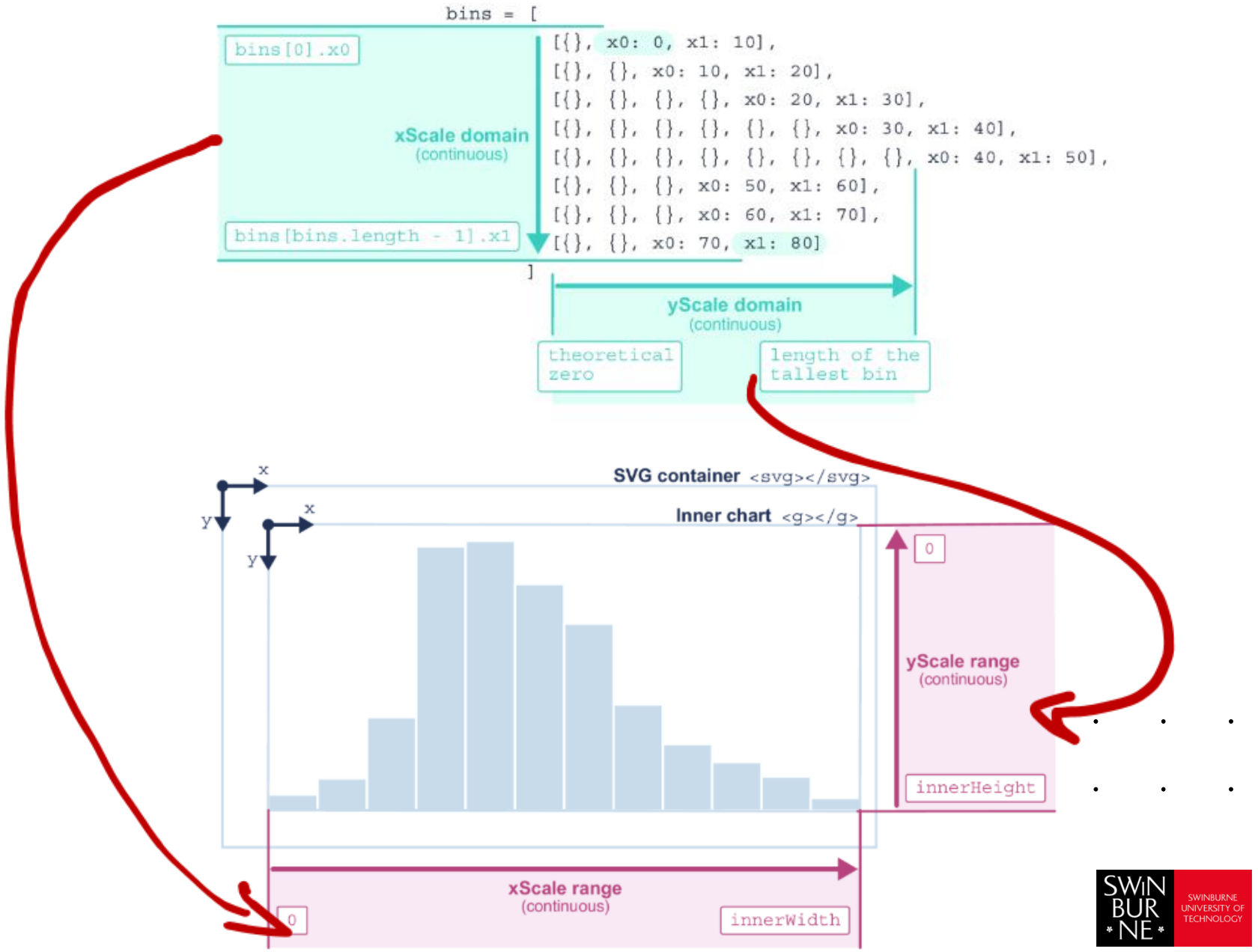
Creating scales

The number of bins

-> xScale domain

The x1 value will give us the height of the bins

-> yScale domain



xScale

```
const minTime = bins[0].x0; //lower bound of the first bin
const maxTime= bins[bins.length - 1].x1; //upper bound of the last bin

const xScale = d3.scaleLinear()
  .domain([minTime, maxTime])
  .range([0, innerWidth]);
```


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yScale

```
const binsMaxLength = d3.max(bins, d => d.length); // Get the maximum length of the bins

const yScale = d3.scaleLinear()
.domain([0, binsMaxLength])
.range([innerHeight, 0])
.nice();
```


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Append rectangles for histogram bars

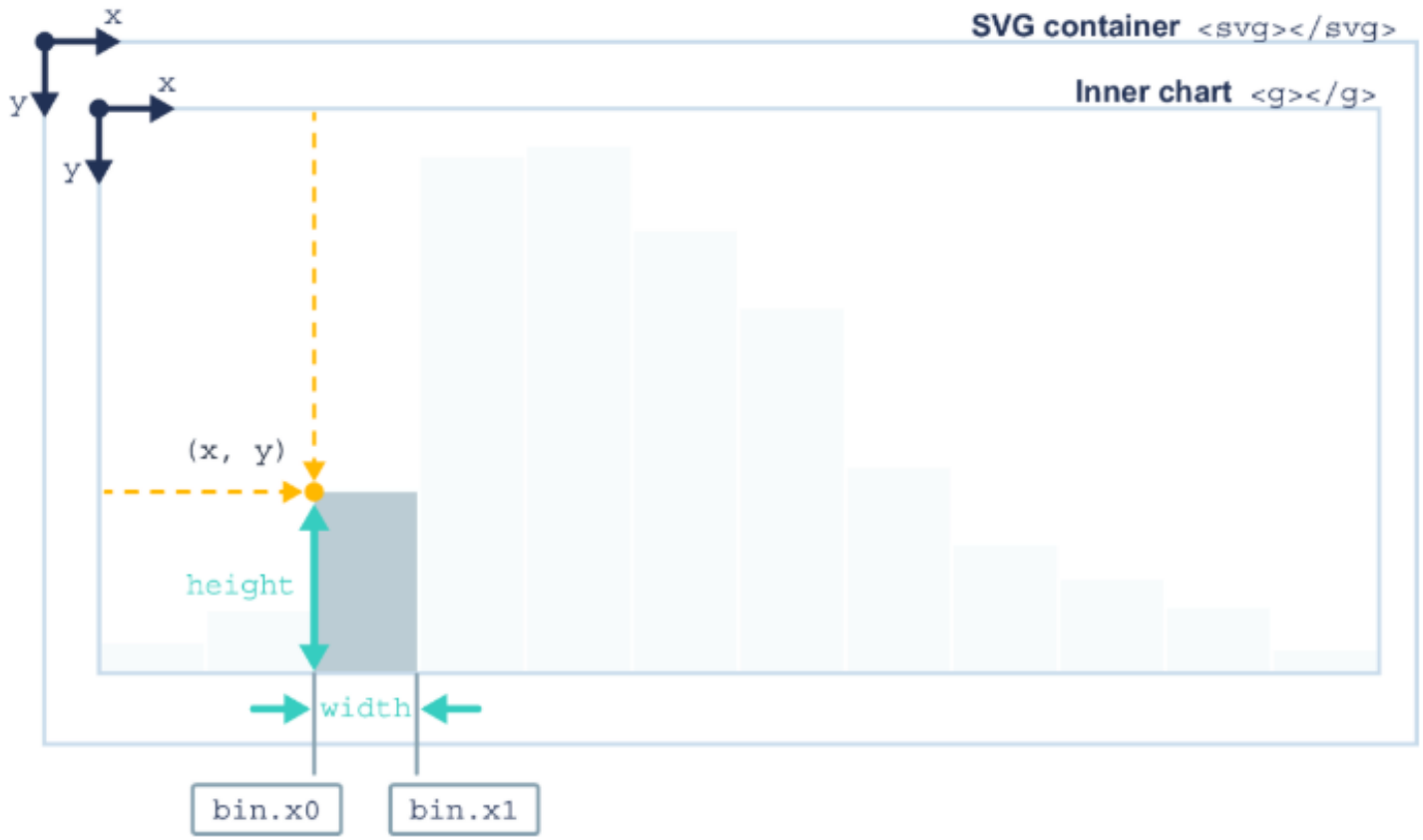
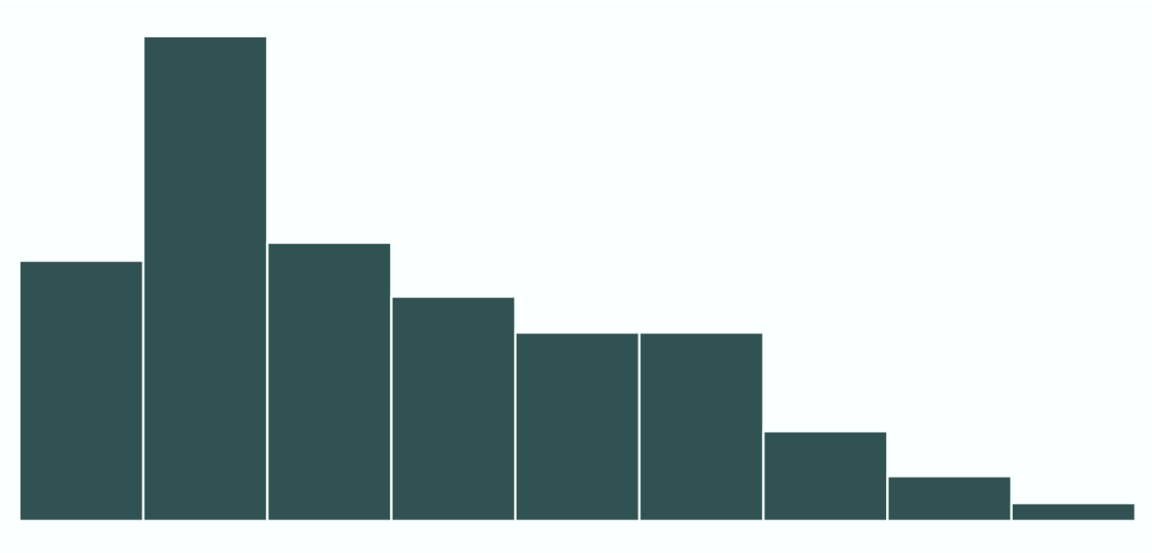
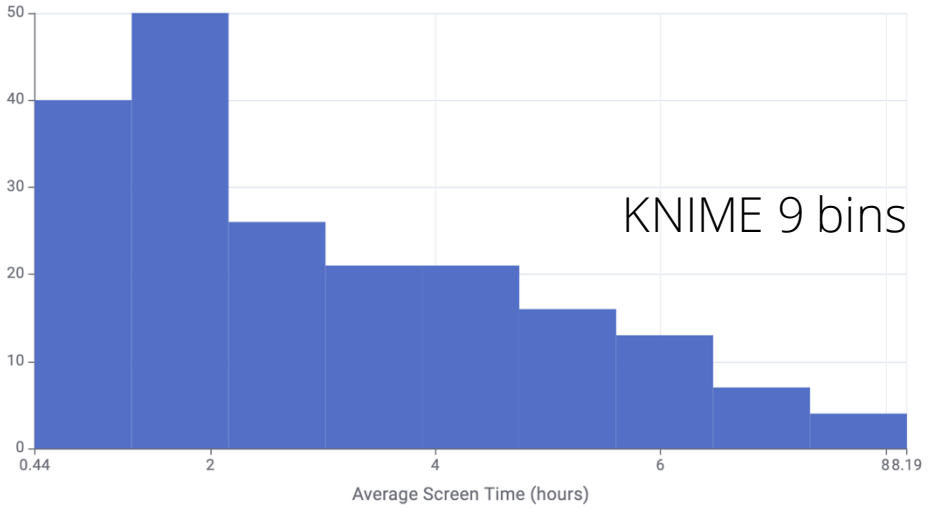


Figure 6.7 The position of the histogram's rectangles is controlled by their `x` and `y` attributes and their dimensions by their `width` and `height` attributes. The height of the rectangles is proportional to the number of data points contained in their related bin.

Drawing histogram

```
innerChart
.selectAll("rect")
.data(bins)
.join("rect")
  .attr("x", d => xScale(d.x0))
  .attr("y", d => yScale(d.length))
  .attr("width", d => xScale(d.x1) - xScale(d.x0))
  .attr("height", d => innerHeight - yScale(d.length))
  .attr("fill", slateGray)
  .attr("stroke", white)
  .attr("stroke-width", 2);
```





KNIME 9 bins

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Adding x-axis

```
// construct the x-axis
const bottomAxis = d3.axisBottom(xScale);


// Add the x-axis to the bottom of the chart relative to the inner chart
innerChart
  .append("g")
  .attr("transform", `translate(0, ${innerHeight})`)
  .call(bottomAxis);
```

```
// Add the x-axis label
svg
.append("text")
.text("Average Screen Time (hours)")
.attr("text-anchor", "end")
.attr("x", width)
.attr("y", height - 5);
```

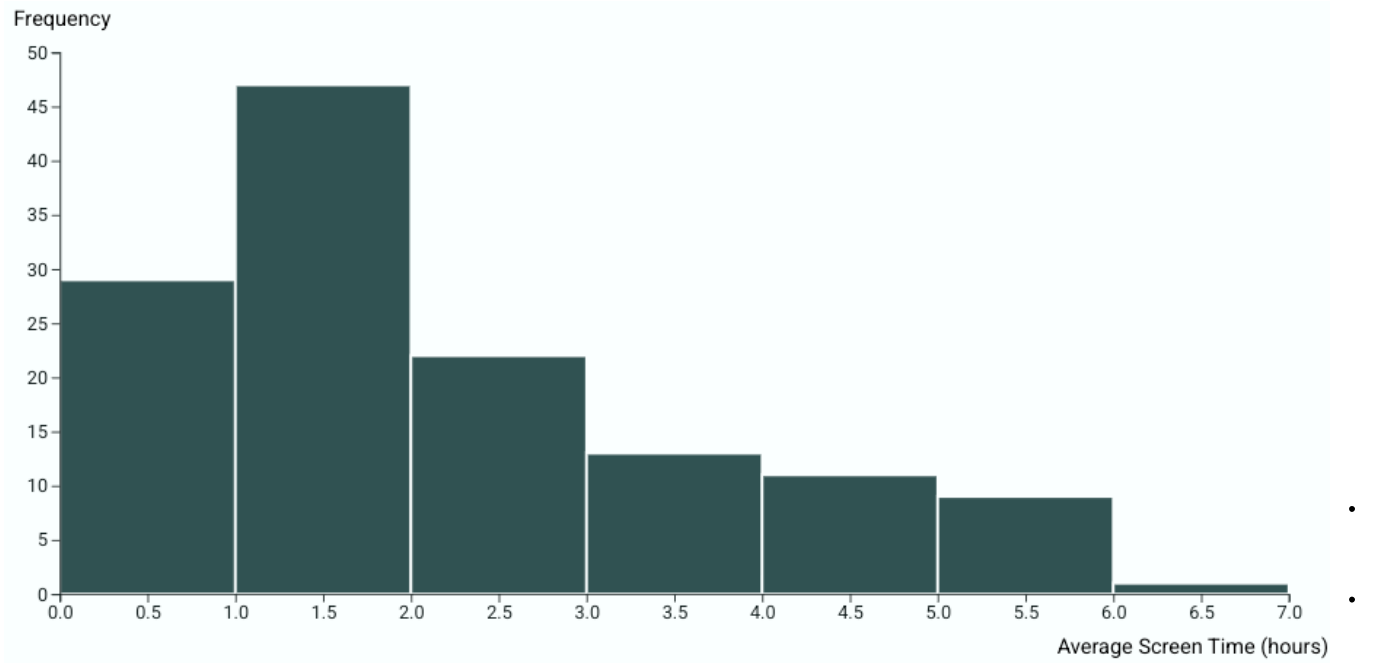

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Adding y-axis

```
// construct the y-axis
const leftAxis = d3.axisLeft(yScale);

// Add the y-axis to the left of the chart relative to the inner chart
innerChart
  .append("g")
  .call(leftAxis);

// Add the y-axis label
        svg
          .append("text")
          .text("Frequency")
          .attr("x", 5)
          .attr("y", 20);
```



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